

Kubernetes & CI/CD Cheat Sheet

kubectl, deployments, services, config, secrets and CI/CD flow.

kubectl Basics

Topic	Quick reference
Context	kubectl config get-contexts; kubectl config use-context NAME
Get resources	kubectl get pods,svc,deploy -n namespace
Describe	kubectl describe pod POD
Logs	kubectl logs -f POD

Deployments

Topic	Quick reference
Apply	kubectl apply -f deployment.yaml
Rollout	kubectl rollout status deployment/app
Restart	kubectl rollout restart deployment/app
Scale	kubectl scale deployment app --replicas=3

Services & Ingress

Topic	Quick reference
Service	ClusterIP for internal, NodePort/LoadBalancer for external
Port forward	kubectl port-forward svc/app 8080:80
Ingress	Route HTTP traffic to services

Config & Secrets

Topic	Quick reference
ConfigMap	kubectl create configmap app-config --from-literal=ENV=prod
Secret	kubectl create secret generic app-secret --from-literal=TOKEN=xxx
Never commit secrets	Use secret managers or encrypted values

CI/CD Flow

Topic	Quick reference
Pipeline stages	Code -> Test -> Build image -> Push registry -> Deploy
Common tools	GitHub Actions, GitLab CI, Jenkins, Azure DevOps
Quality gates	lint, unit tests, security scan, approval gates